Research Supplement S1–S8

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Supplement to “Calibrated Strict Avalanche Criterion Comparisons Reveal Detector-Dependent Diffusion Endpoints”

This index accompanies the revised manuscript. A standalone PDF provides the same supplemental narrative and tables for journal upload. Experimental sources and data retain their original hashes and execution commits. This document consolidates definitions, complete contrast reporting and evidence locations; it does not add cipher replications or replace historical outcomes. Links identify artifacts within the repository; a pinned full-history clone supports exact reproduction.

S1. Observation model, search layouts and auxiliary decision bounds

The primary archive is reference data. Bit zero is the integer LSB. RECTANGLE rows occupy bits 0–15,16–31,32–47,48–63; CLEFIA words are packed most-significant first; SPARKLE and Alzette words are packed least-significant first. The methods are defined by the retained adapters, not by a renderer's byte order. Input stars contain a uniform base plus all n one-bit neighbours. Trace queries include key additions/whitening as documented; final ciphertext finalization is checked separately.

Active input sets: RECTANGLE {0,16,32,48} and {0,16,32,48,1,17,33,49} select one/two bit-slice columns; CLEFIA {88,…,91}/{88,…,95} select four/eight bits of plaintext byte 4 (byte 0 is most significant); SPARKLE/Alzette use {0,…,3}/{0,…,7} in the first low word. These fixed, simple layouts are reproducible restricted probes, not optimized or exhaustive searches. Three differential patterns flip the first active bit, all four active bits or all eight active bits.

Under independent output labels at distinct inputs, conditional on a base label the neighbour XORs are independent uniform n-bit vectors, with a law that does not depend on the base label. Disjoint stars therefore give iid Bin(N,1/2) count cells. Star intersections have probability at most C(N,2)(n+1)^2/2^n; permutation switching adds C(Q,2)/2^n, Q=N(n+1). Their sum is bounded by Q^2/2^n. Coupling bounds changes of event probability, so the capped bound is added to ideal-label test tails. This proves a reference-model correction, not independence of actual cipher coordinates or serial round outcomes.

For linear sign agreements use two-sided exact Bin(N,1/2) cell tails, Bonferroni over n^2 single-bit input/output masks, and delta_L=min(1,N^2/2^n). For K distinct k-bit input cosets, each output-coordinate XOR is fair under labels. If M coordinates are zero in every coset, use Pr[Bin(n,2^-K)>=M] + C(K,2)/2^(n-k) + C(K*2^k,2)/2^n, capped at one. The correction includes duplicate cosets and label collisions. A conservative alternative is n*2^-K/M plus the same correction (M>0). The archive calls these ZS4/ZS8; “coset XOR-balance” identifies the tested coordinate property more precisely than a full-state zero-sum construction.

Differential detection retains T>=3, where T is maximum full-output difference multiplicity over three fixed differences, and requires D*[C(N,3)/((2^n-3)(2^n-5)) + C(N,2)*2/2^n] < alpha, D=3. For a fixed nonzero input difference, distinct unordered input pairs are disjoint; after a first output pair fixes a difference, the next two pairs match it with probabilities bounded by 1/(2^n-3) and 1/(2^n-5). Union bounds cover all triples and D patterns. The second term covers repeated unordered input pairs. The bound is for the event T>=3, not a tail estimate indexed by every possible observed T. A T>=2 detector would need its own derivation, frozen policy and multiplicity handling; it was not substituted.

The full derivations and historical policies remain in METHODOLOGY_ID.md and METHODS.md. Finalization, output entropy and Python timings are ancillary audits rather than evidence establishing diffusion security.

S2. Original horizon components and historical SAC comparison

The complete seed/key/policy component table is horizon_semua_aturan.csv. “margin_ronde” in historical summaries means the remaining observed window R-H, not a cryptanalytic security margin. A dash/empty family horizon means H_m=0. Alzette H=R=4 is right-censored. Slim shares big prefixes.

All seven historical Stage-B significant contrasts are shown below. The historical policy is MC-calibrated histograms at alpha=0.001; Count uses its corrected discrete rule. These results have unequal actual null sizes and are not silently replaced by Stage C. Δ = comparator minus Count; b=Count-only, c=comparator-only. Intervals are pointwise paired bootstrap, not simultaneous post-selection coverage. Exact p-values remain in the source CSV.

r	N	Comparator	b	c	Δ (pp)	95% CI (pp)	p-Holm90
6	64	Basic	0	271	54.2	[49.8, 58.4]	4.64×10^{-80}
6	64	Row	45	204	31.8	[26.2, 37.2]	1.98×10^{-23}
6	64	Column	80	159	15.8	[9.8, 21.6]	3.04×10^{-5}
7	64	Basic	0	112	22.4	[18.8, 26.2]	3.35×10^{-32}
7	256	Basic	0	486	97.2	[95.6, 98.6]	9.01×10^{-145}
7	1,024	Basic	0	439	87.8	[84.8, 90.6]	1.25×10^{-130}
7	1,024	Column	54	19	−7.0	[−10.4, −3.8]	4.25×10^{-3}

The full two-cipher 30-condition grid, all 90 primary contrasts, Wilson intervals, paired outcomes and 64-/128-bit null validation are in Stage-B data and its protocol. Both state widths are retained, including the historical 128-bit,N=256 Basic null rate 0.00123. Original coset detail, calibration and larger-sample results remain in the reference folders.

S3. Size-aware reanalysis and residual null mismatch

Three independent million-draw banks per N and one independent million-draw validation per N give nine million training and three million held-out draws. The Count-size-matched policy (archive identifier `match_count_nonrandom`) is primary; randomized α = 0.001 (`equal_alpha_randomized`) is sensitivity. Bank zero is primary and other banks measure retraining sensitivity. Validation is shared across banks and never used to replace bank zero. Preserving nonrandom Count targets its actual discrete size, but does not force each trained histogram rule to have exactly that conditional size. The randomized policy is separate and explicitly changes all detectors.

All six significant primary reanalysis contrasts follow (same definitions as S2). Holm90 retains Stage B's conservative family-size parameter M = 90 for the 45 ordered RECTANGLE p-values, with multipliers 90 − j + 1 and a running maximum capped at one. Historical Column r6,N64 significance disappears after recalibration. The full 45 tested rows, including non-significant results, are available in the linked data.

r	N	Comparator	b	c	Δ (pp)	95% CI (pp)	p-Holm90
6	64	Basic	0	271	54.2	[49.8, 58.4]	4.64×10^{-80}
6	64	Row	74	169	19.0	[13.2, 24.8]	8.70×10^{-8}
7	64	Basic	0	79	15.8	[12.6, 19.0]	2.88×10^{-22}
7	256	Basic	0	477	95.4	[93.6, 97.2]	4.61×10^{-142}
7	1,024	Basic	0	439	87.8	[85.0, 90.6]	1.25×10^{-130}
7	1,024	Column	55	18	−7.4	[−10.6, −4.2]	1.44×10^{-3}

Bank-zero conditional null validation (each row has one million draws) is shown for both primary histogram laws. Row and Column share the marginal maximum-axis law, not joint decisions. Count null draws are simulated separately; no cross-detector paired-null claim is made.

N	Law	Count target	Observed null size	Wilson95 interval
64	Basic	0.000301282	0.000271	[0.000240602, 0.000305237]
64	Maximum-axis	0.000301282	0.000272	[0.000241543, 0.000306297]
256	Basic	0.000406803	0.000346	[0.000311418, 0.000384421]

N	Law	Count target	Observed null size	Wilson95 interval
256	Maximum-axis	0.000406803	0.000374	[0.000337974, 0.000413865]
1,024	Basic	0.000718038	0.0007	[0.000650045, 0.000753791]
1,024	Maximum-axis	0.000718038	0.000775	[0.000722342, 0.000831493]

All banks/policies and exact statistics are in size-matching evidence, interpretation and frozen protocol/reproduction. Upper tails retain 5,000 order statistics, validation indices and count boundary candidates. Full-array hashes and execution-time full/tail decision checks include truncated tie groups. The portable verifier preserves frozen source bytes and adapts only one floating-point target metadata comparison; scientific decisions and integer/hash checks remain unchanged.

S4. Independent inputs and limited random-key confirmation

The plan and full key list precede main measurements. Display keys K1–K8 map to archived identifiers random_01–random_08; Default maps to default. Nine keys comprise a default control and eight randomly selected keys excluding historical keys; all are retained irrespective of performance. Each key has 500 replications, N=256/1024 and rounds 6–8. The 18 preselected contrasts have Holm18 control; the complete grid has separate Holm162 control. No 4,000-input-replication pooling is used to claim 4,000 independent keys. Intervals condition on key and trained calibration; different keys have different input samples.

Key	N	Comparator	Δ (pp)	95% CI (pp)	p-Holm18	p-Holm162
Default	256	Basic	95.4	[93.4, 97.2]	7.69×10^{-143}	8.15×10^{-142}
Default	1,024	Column	−11.6	[−15.0, −8.2]	1.30×10^{-9}	2.08×10^{-8}
K1	256	Basic	96.6	[95.0, 98.2]	1.36×10^{-144}	1.29×10^{-143}
K1	1,024	Column	−3.0	[−6.2, 0.2]	0.0912	1
K2	256	Basic	95.4	[93.4, 97.2]	7.69×10^{-143}	8.15×10^{-142}
K2	1,024	Column	−10.8	[−14.4, −7.2]	4.12×10^{-8}	7.37×10^{-7}
K3	256	Basic	97.0	[95.4, 98.4]	3.60×10^{-145}	3.24×10^{-144}
K3	1,024	Column	−7.8	[−11.4, −4.2]	2.66×10^{-4}	7.44×10^{-3}
K4	256	Basic	95.4	[93.4, 97.2]	7.69×10^{-143}	8.15×10^{-142}
K4	1,024	Column	−6.2	[−9.8, −2.6]	3.05×10^{-3}	0.139
K5	256	Basic	94.6	[92.4, 96.6]	9.76×10^{-140}	1.50×10^{-138}
K5	1,024	Column	−8.8	[−12.2, −5.2]	1.11×10^{-5}	2.25×10^{-4}
K6	256	Basic	95.2	[93.2, 97.0]	1.23×10^{-142}	1.60×10^{-141}
K6	1,024	Column	−6.6	[−9.8, −3.4]	5.67×10^{-4}	0.0197
K7	256	Basic	95.6	[93.6, 97.2]	4.10×10^{-143}	4.10×10^{-142}
K7	1,024	Column	−5.4	[−8.8, −2.0]	6.37×10^{-3}	0.433
K8	256	Basic	94.8	[92.8, 96.6]	4.51×10^{-142}	6.36×10^{-141}
K8	1,024	Column	−8.2	[−11.6, −4.8]	3.39×10^{-5}	7.98×10^{-4}

See all key evidence, frozen keys/protocol, and confirmation interpretation. Zero duplicate/intersecting bases concern bases only, not all neighbour queries. Raw statistics are reconstructed from full value and axis-bin histograms; 72 native eight-round snapshot traces are regenerated. Selected complete snapshots do not imply every full matrix was originally retained.

S5. Exploratory mechanism diagnostics

All 54 key/N/round metrics, 432 bin contributions, PMFs, coordinate maps, surrogate rates, covariance decomposition and parent-clustered layout results are in mechanism data. SAC_MECHANISM.md explains them with four diagnostic figures; the frozen protocol defines every seed and fit.

Mean Pearson equals histogram shape plus replication scatter exactly, using the divisor L rather than L-1 for the descriptive variance. The empirical independent scatter is sum(pi_ijb*(1-pi_ijb)/E_b); actual-minus-independent scatter is signed net covariance. A small net term neither proves independent coordinates nor quantifies how much detection power is explained.

For pooled coordinate totals T over M independent bases, unbiased squared bias is ((T-M/2)^2-M/4)/(M*(M-1)). Negative estimates are retained; averaging precedes the nonnegative square-root display. Pooled SAC marginal tests have a fixed-map probability-1/2 null and a 221,184-coordinate Bonferroni family. That family does not cover every adaptive project analysis. They use 500 times one replication's budget and must not be presented as per-replication ideal-permutation detection or perfect-SAC conclusions.

Three default cases are replayed from the existing inputs: N256/r7, N256/r8, N1024/r7. All 3,000 original analyzed-round hashes are checked. Each surrogate fits on 250 existing parents and is diagnosed against the other 250. There are 90,000 synthetic matrices and 30,000 spatial permutations, not 120,000 new cipher replications. Product-marginal models remove all cross-cell dependence; they do not isolate a shared-base causal mechanism. Each of 250 parent matrices has 40 permutations; bootstrap resamples parents. Rare parent outcomes and omitted fit uncertainty limit intervals. The iid-fair 10,000-draw rate is a diagnostic, not a replacement for million-draw null validation.

Additional mechanism figures

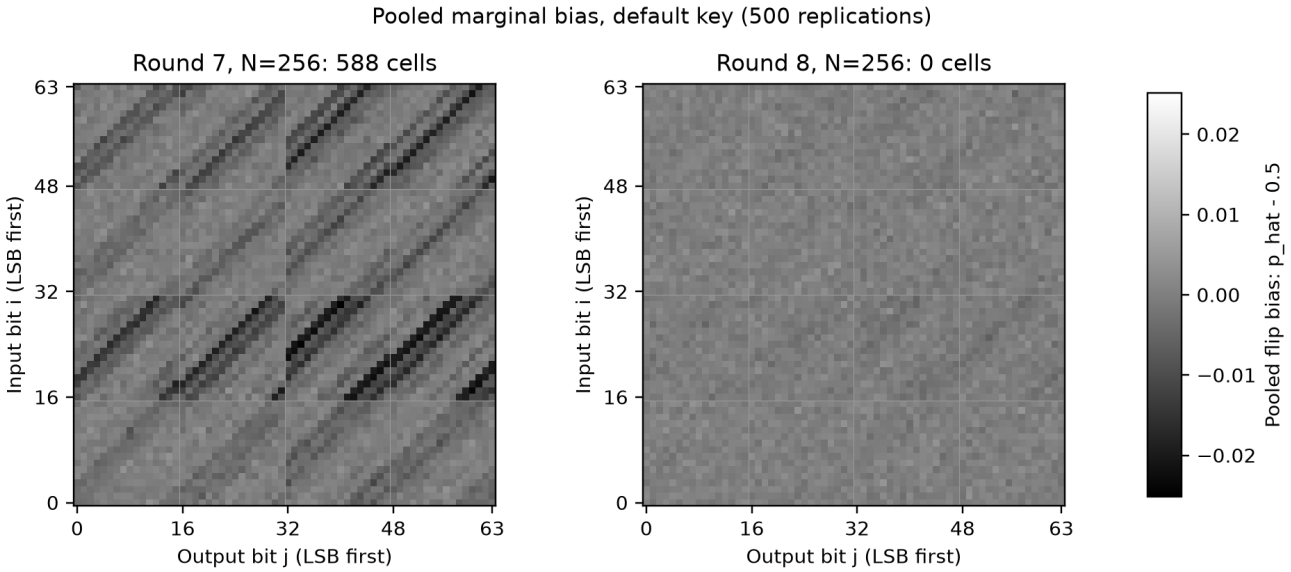


Figure S1. Default-key pooled marginal flip-probability deviations at rounds 7/8, N = 256. The pooled budget and marginal null differ from a single detector replication.

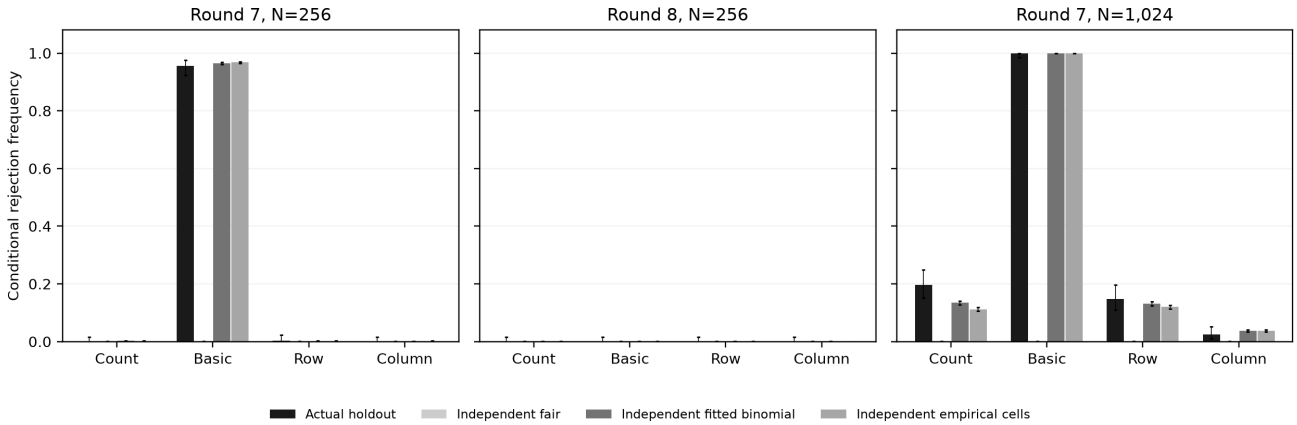


Figure S2. Default-key surrogate diagnostics. Fitted models condition on the first 250 replications; intervals omit parameter-fitting uncertainty.

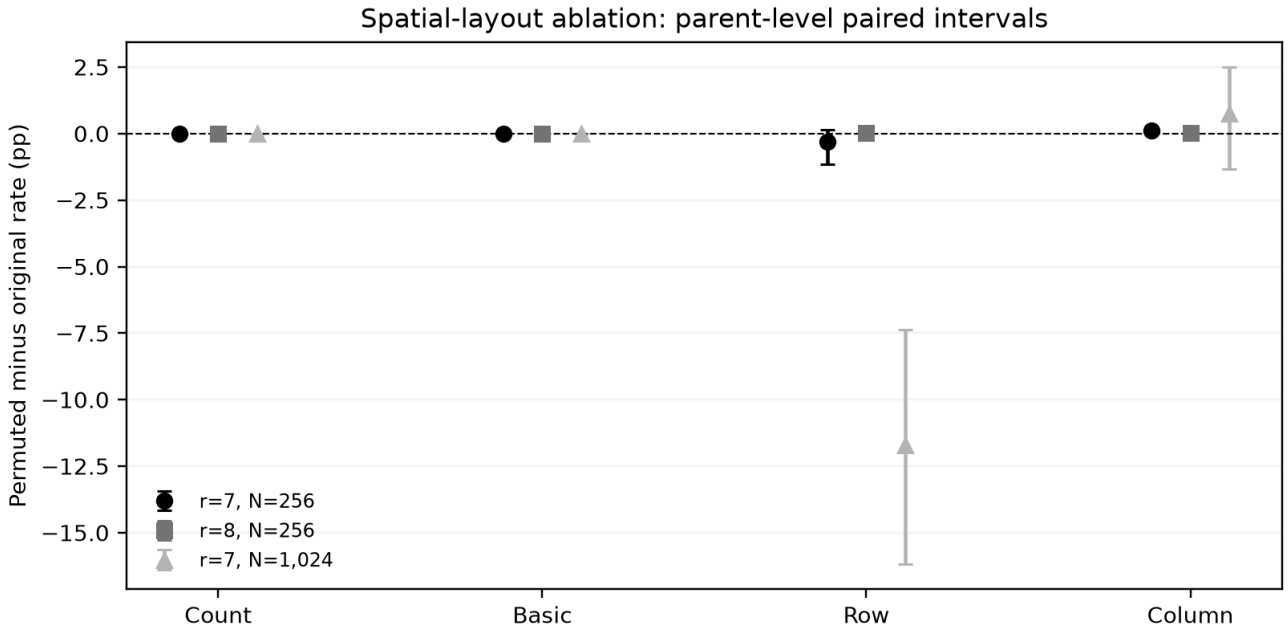


Figure S3. Cell-position permutation diagnostics. The 40 rearrangements per parent are clustered; intervals resample 250 parent matrices.

S6. Late decisions and ancillary audits

Across six original runs, six base traces and five corrected families, after the corresponding combined H, excluding slim and the paired-CLEFIA view, there are 1,770 decisions and three rejected positions: CLEFIA r13/r14 in the larger run and SPARKLE256 step10 at seed2028. The CLEFIA pair is one carried observable. Neither a naive count-times-alpha expectation nor failure of confirmation identifies a particular outcome as a proven false positive. The SPARKLE late signal has no dedicated confirmation.

The selected CLEFIA observable loses confirmation on 65,536 new inputs. See reference audit files for the frozen coordinates, confirmation and late decisions. Tie-aware Spearman, within-primitive/without-slim descriptions, whitening/output entropy, key-avalanche and scalar-Python timings are ancillary material retained in the reference archive. Constant-series correlations are undefined and pooled serial observations are not independent statistical evidence. Uniform inputs stay uniform even under identity; output uniformity alone is not diffusion.

S7. Implementation, provenance and reproduction

Official RECTANGLE/CLEFIA vectors and native scalar/batch checks are recorded in regression tests. Official SPARKLE C v1.2.1 conformance, retained source, license and permanent reports are indexed by SPARKLE_C_VALIDATION.md. Four builds reuse inputs; slim shares prefixes. Finite agreement is not exhaustive conformance or independently developed statistical replication. The old blocked Windows attempt remains recorded separately from successful Linux execution.

REPRODUCING.md and experiment READMEs give commands and pinned dependencies. A full-history clone is required for source-commit byte checks. The four revised-manuscript figures can be regenerated with `python -B experiments/plot_manuscript_figures.py` after installing `experiments/sac_confirm/requirements.txt`. PNGs and script/input/output hashes are written under ignored `results/manuscript-figures/`; no measurements are changed. The reproducer emits script/input/output hashes for all four figures. CI checks manifest integrity, regression, C conformance and retained SAC inference, including native snapshots and mechanism diagnostics. It does not regenerate all expensive training arrays or 4,500 full input replications. SHA256SUMS.json protects intended repository files. Original/follow-up protocol and execution commits are preserved.

The standalone PDF is rendered from this index and the linked diagnostic PNGs with `scripts/render_supplement_pdf.py --fonts /path/to/Times-fonts`. This document-only command requires ReportLab and locally supplied Times New Roman font files; fonts are not distributed. It leaves all numerical measurements unchanged.

Substantive AI assistance, validations and recorded model-identifier limits are disclosed in AI_ASSISTANCE.md.

S8. Reporting checklist and scope of claims

For each empirical endpoint record:

1 Primitive/variant, native iteration unit and observed window.

2 State packing, finalization, key selection and access interface.

3 Observable, statistic, detector family and fixed search layout.

4 Base count, total trace-query budget and reuse/nesting.

5 Null model, finite-domain correction and measured conditional size.

6 Calibration training/validation separation and retraining sensitivity.

7 Multiplicity family, paired inference and conditional interval scope.

8 Endpoint definition, component support, late outcomes and censoring.

9 Selection versus new-input/key confirmation versus reused diagnostics.

10 Immutable repository version, source/data hashes and reproduction commands.

No optimal detector, causal cipher-layer attribution, new contiguous Basic horizon, whole-key-space equivalence or full-round security claim is made.